

ADVANCED QUANTUM MECHANICS

Exercise 6 – 18.11.2025



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Problem 1 – harmonic oscillator in a uniform electric field

Consider a one-dimensional charge q of mass m in a harmonic trap of frequency ω , subject to a static electric field E along x :

$$\hat{H} = \hat{H}_0 + \hat{V}, \quad \hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \quad \hat{V} = -q\mathcal{E}\hat{x}.$$

- (i) Using the SHO eigenbasis $\{|n\rangle\}$ of $\hat{H}_0$, compute $E_n^{(1)} = \langle n | \hat{V} | n \rangle$ and show it vanishes.
- (ii) Show that the only nonzero matrix elements are between $|n\rangle$ and $|n \pm 1\rangle$, and derive

$$E_n^{(2)} = -\frac{(qE)^2}{2m\omega^2}.$$

- (iii) Complete the square to show the potential minimum is displaced by

$$x_0 = \frac{qE}{m\omega^2},$$

and that $\hat{H}$ is unitarily equivalent to $\hat{H}_0$ up to a constant shift. Comment on the convergence of perturbation theory.

Hint. Use $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger)$ and $\langle n | a | n+1 \rangle = \sqrt{n+1}$, $\langle n | a^\dagger | n-1 \rangle = \sqrt{n}$.

Problem 2 – weak-field Zeeman effect

Consider the hydrogen atom in a weak magnetic field, ignoring the electron and nuclear spin. The Zeeman shift is then described by the Hamiltonian

$$\hat{H}' = \mu_B B \hat{L}_z / \hbar, \quad \mu_B = \frac{e\hbar}{2m_e}.$$

- (i) Calculate the first order shift

$$\Delta E_{n\ell m}^{(1)} = \langle n\ell m | \hat{H}' | n\ell m \rangle.$$

Comment on the lifting of the m -degeneracy for a given ℓ .

- (ii) List the levels $\{3s, 3p, 3d\}$ with ℓ and their m values and sketch the linear splitting vs. B . Which levels remain unshifted in this orbital-only model?

Problem 3 – finite square well and tunneling

Consider the symmetric finite square well

$$V(x) = \begin{cases} -V_0, & |x| < a, \\ 0, & |x| \geq a, \end{cases} \quad V_0 > 0.$$

- (i) For energies $-V_0 < E < 0$, write the general solutions inside/outside:

$$\psi(x) = \begin{cases} A \cos(kx) + B \sin(kx), & |x| < a, \quad k = \sqrt{\frac{2m(E+V_0)}{\hbar^2}}, \\ C e^{-\kappa|x|}, & |x| \geq a, \quad \kappa = \sqrt{\frac{-2mE}{\hbar^2}}. \end{cases}$$

Use parity to split into even/odd solutions.

- (ii) Impose continuity of ψ and ψ' at $x = \pm a$ to derive the transcendental conditions

$$\text{even: } k \tan(ka) = \kappa, \quad \text{odd: } -k \cot(ka) = \kappa.$$

- (iii) Show that in the limit $V_0 \rightarrow \infty$ these reduce to the infinite-well conditions $ka = n\pi/2$ (with appropriate parity). Give the criterion for how many bound states exist as a function of the dimensionless depth $z_0 \equiv a\sqrt{2mV_0}/\hbar$.
- (iv) For $z_0 = 5$, sketch the LHS/RHS of the even/odd equations versus ka and locate the intersections.
- (v) Replace the well by a barrier $V(x) = V_0$ for $|x| < a$ and 0 outside. For $E < V_0$, write the transmission probability T through one barrier (no multiple reflections) and show its dominant dependence

$$T \sim \exp\left(-2a \sqrt{\frac{2m(V_0 - E)}{\hbar^2}}\right).$$